

Account Restriction Notice
Coupons
Get Coupon Here
Country / Region
Language
undefined
Currency
USD \$

The currency you select is for reference only, payments must be made in US dollars.

Language Coming Soon
Ship to
JLCPCB
Products

PCB Service

- FR-4 PCBs
- Flexible PCBs
- Metal Core PCBs
- High-Frequency PCBs

PCBA Service

- PCB Assembly
- Parts Library

More

- SMT Stencil
- PCB Layout

Mechanical Services

- JLC3DP - 3D Printing
- JLCCNC - CNC Machining
- JLCCNC - Sheet Metal
- JLCMC - Mechanical Components

Tools

- EasyEDA - PCB Design Tool
- JLCDFM - DFM Check
- Impedance Calculator
- PI Stiffener Calculator

More

- OSHWLAB - Open Source Hardware
- JLCPCB - Flexible Heater

Capabilities Support

Contact Us

- Online Chat>
Mon-Fri: 24 hours, Sat: 9am-6pm, GMT+8
- Email Us>
Mon-Fri: 24 hours, Sat: 9am-6pm, GMT+8
- WhatsApp>
Mon-Fri: 24 hours, Sat: 9am-6pm, GMT+8

Support

- Help Center
Get instant answers. 24/7 Available.
- Blog
- Community

About Us
About Us

JLCPCB delivers one-stop electronic and mechanical manufacturing solutions to empower hardware innovators worldwide.

Explore Us

- Why JLCPCB?
- Quality Assurance
- How We Work
- Certifications
- Company News

Search

Order Now

Sign In Sign In

Rigid PCB Manufacturing Capabilities

JLCPCB offers 1-32 layer rigid PCB manufacturing and assembly services with various substrate thicknesses, surface finishes, and impedance control.

Green
Purple
Red
Yellow
Blue
White
Black

Rigid PCB Capabilities

Flex PCB Capabilities
PCBA Capabilities
Stencil Capabilities
PCB Layout Capabilities
Flexible Heaters Capabilities

PCB Specifications

Features	Capability	Description	Patterns
Layer count	1-32 Layers	The number of copper layers in the PCB	
Controlled Impedance	4/6/8/10/12/14/16/18/20/.../32 layers	User Guide to the JLCPCB Impedance CalculatorJLCPCB Impedance Calculator	
Impedance Tolerance	±10%		
Material	FR-4	Grade A laminates from suppliers including Nan Ya, KB, Shengyi and etc.	
	Aluminum-Core	1-layer Aluminum-core PCBs	
	Copper-Core	1-layer copper-core PCBs with direct heatsink contacts to core (≥ 1 × 1 mm)	
	RF PCB	1 oz copper, 2-layer RF PCBs with Rogers and PTFE cores	
FR-4 Dielectric Constants	4.5 (2-Layer PCB)	7628 Prepreg 4.4 3313 Perpreg 4.1 2116 Perpreg 4.16	
Maximum Dimensions	FR4(1-layer): 606 × 510 mm FR4(2-layer): 670 × 600 mm FR4(4-layer): 663 × 593 mm FR4(6-layer and above): 656 × 586 mm Rogers / PTFE Teflon PCB: 590 × 438 mm Aluminum PCB: 602 × 506 mm Copper PCB: 480 × 286 mm		These limits apply to PCBs with thickness ≥ 0.8 mm. The thinner FR4 PCBs are 599 × 497 mm maximum. 2-layer FR4 PCBs can reach a maximum size of 1020 × 600 mm. 4-layer FR4 PCBs can reach a maximum size of 1016 × 596 mm.

Minimum Dimensions	FR4/Rogers/PTFE: 3 × 3 mm Castellated / Plated Edges: 10 × 10 mm Aluminum/Copper PCB: 5 × 5 mm	These limits apply to PCBs with thickness ≥ 0.6 mm. Manual review required for thinner PCBs. Panelization is recommended for small-sized boards.
Dimension Tolerance	±0.1mm	±0.1mm(Precision) and ±0.2mm(Regular) for CNC routing, and ±0.4mm for V-scoring
Thickness	0.4 – 4.5 mm	Thickness for FR4 are: 0.4/0.6/0.8/1.0/1.2/1.6/2.0 mm (2.5 mm and above are for 12+ layer PCBs only)
Thickness Tolerance (Thickness≥1.0mm)	± 10%	e.g. For the 1.6mm board thickness, the finished board thickness ranges from 1.44mm(T-1.6×10%) to 1.76mm(T+1.6×10%)
Thickness Tolerance (Thickness<1.0mm)	± 0.1mm	e.g. For the 0.8mm board thickness, the finished board thickness ranges from 0.7mm(T-0.1) to 0.9mm(T+0.1).
Finished Outer Layer Copper	2-layer: 1 oz / 2 oz / 2.5 oz / 3.5 oz / 4.5oz Multi-layer: 1 oz / 2 oz	
Finished Inner Layer Copper	0.5 oz / 1 oz / 2 oz	Finished copper weight of inner layer is 0.5oz by default.
Soldermask	Green, Purple, Red, Yellow, Blue, White, and Black.	We use LPI (Liquid Photo Imageable) solder mask. This is the most common type of mask used today. Heat-cured ink soldermask is usually found on low-cost, single-sided PCBs.
Surface Finish	HASL (leadless / lead-free), ENIG, OSP (Single-sided FR-4, FPC, and aluminum PCBs are currently not supported for OSP)	Aluminum PCBs: HASL only; FR-4/HDI boards with 6 or more layers, boards with a thickness ≤ 0.4 mm, high-frequency boards, copper-based PCBs, and FPCs do not support HASL.

Drilling

Features	Capability	Description	Patterns
Drill Diameter	1-layer: 0.3 – 6.3 mm 2-layer: 0.15 – 6.3 mm Multilayer: 0.15 – 6.3 mm	Holes with diameter ≥ 6.3 mm are CNC routed from a smaller drilled hole. Min. drill diameter for 2- or more-layer PCBs is 0.15 mm (more costly!) Min. drill diameter for aluminum-core PCBs is 0.65 mm Min. drill diameter for copper-core PCBs is 1.0 mm	
Hole size Tolerance	Through-holes: +0.13 / -0.08 mm Press-fit holes : ±0.05 mm (Finished hole size: 0.55-2.0mm. Multilayer ENIG boards and circular holes only. Mention the specific holes in PCB Remark)	e.g. for the 0.6mm hole size, the finished hole size between 0.52mm to 0.73mm is acceptable. To avoid soldermask or tin stuck in the hole, recommended PTH size≥0.5mm.	
Average Hole Plating Thickness	18µm		
Hole Position Tolerance	±0.05 mm		
Blind/Buried Vias	Not supported	Currently we don't support Blind/Buried Vias, only make through holes.	

Min. Via hole size/diameter	0.15mm / 0.25mm	1-layer (NPTH only): 0.3 mm hole size / 0.5 mm via diameter 2-layer: 0.15mm hole size / 0.25mm via diameter Multilayer: 0.15 mm hole size / 0.25 mm via diameter ① Via diameter should be 0.1mm(0.15mm preferred) larger than Via hole size. ② Preferred Min. Via hole size: 0.2mm ③ 0.15mm hole size with any size via diameter, and 0.2mm or 0.25mm hole size with via diameter less than 0.45mm, will cost more. Please select corresponding via size option when placing order.
Min. Non-plated holes	0.50mm	Please draw NPTHs in the mechanical layer or keep out layer.
Min. Plated Slots Width	2-layer: 0.5mm Multi-layer: 0.35mm	The minimum plated slot width is 0.5mm, which is drawn with a pad. The length of the slot should be at least 2 times of the width.
Min. Non-Plated Slots	1.0mm	The minimum Non-Plated Slot Width is 1.0mm, please draw the slot outline in the mechanical layer(GM1 or GKO)
Slot Hole size Tolerance	Plated: +0.13 / -0.08 mm Non-plated: ±0.2mm	Plated slot is made by drill tools. Non-plated slot is made by CNC.
Via Hole-to-Hole Spacing	0.2mm	
Pad Hole-to-Hole Spacing	0.45mm	
Min. Castellated Holes	0.5mm	Castellated holes are metalized half-holes on PCB edges, commonly used on daughter boards to be soldered onto carrier PCBs. ① Hole diameter (Φ): ≥ 0.5 mm ② Hole to board edge (L): ≥ 1 mm ③ Hole to hole (D): ≥ 0.5 mm ④ Min. PCB size : 10 × 10 mm ⑤ Min. PCB thickness: 0.6 mm
Plated Edges	10 x 10mm	Plated edges are copper-plated and ENIG treated. HASL is not supported. ① Min. PCB size: 10 × 10 mm ② Min. PCB thickness: 0.6 mm ③ At least 3 breaks (more for larger PCBs) in the edge plating are required for support tab connections
Blind Slot		① Blind slot width (W): ≥ 1.0mm ② Blind slot depth (D): ≥ 0.2mm ③ Blind slot annular width (A): ≥ 0.3mm (The pad width of PTH blind slots) ④ Safety distance (S): ≥ 0.2mm (The distance from NPTH blind slots to pad/traces/copper plane) ⑤ Blind slot remaining thickness (R): ≥ 0.2mm (The distance from the bottom of the blind slot to the nearest inner copper layer/surface substrate) ⑥ Supports 2-32-layer FR4 boards with a thickness of ≥ 0.8mm

Backdrill		Backdrill uses a secondary drilling process to control the depth of the hole, removing excess copper from the other layers of the via, thereby reducing its interference with the signal. ① Supports 4-32-layer FR4 boards with a thickness of $\geq 0.8\text{mm}$ ② Through-Hole Diameter(D): 0.2-0.5mm(Epoxy filled after backdrilling) ③ Backdrill Diameter(W): typically 0.2mm larger than through-hole diameter ④ Backdrill Depth(L): layers with backdrilling, customizable ⑤ Dielectric Thickness(T): $\geq 0.15\text{mm}$(from bottom of backdrill to the adjacent inner copper layer) ⑥ Safety Distance(S): $\geq 0.2\text{mm}$(edge to pad/trace/copper safety distance)
Rectangular Holes / Slots	Not supported	Rectangular holes and slots without rounded corners are not supported.

Traces

Features	Capability	Description	Patterns
Min. track width and spacing (1 oz)	0.10 / 0.10 mm (4 / 4 mil)		1- and 2-layer: 0.10 / 0.10 mm (4 / 4 mil) Multilayer: 0.09 / 0.09 mm (3.5 / 3.5 mil). 3 mil is acceptable in BGA fan-outs.
Min. track width and spacing (2 oz)	0.16 / 0.16 mm (6.5 / 6.5 mil)		2-layer: 0.16 / 0.16 mm (6.5 / 6.5 mil) Multilayer: 0.15 / 0.15 mm (6 / 6 mil)
Min. track width and spacing (2.5 oz)	2 layer: 0.2/0.2mm(8mil/8mil)		
Min. track width and spacing (3.5 oz)	2 layer: 0.25/0.25mm(10mil/10mil)		
Min. track width and spacing (4.5 oz)	2 layer: 0.3/0.3mm(12mil/12mil)		
Track width tolerance	$\pm 20\%$	e.g. For a 0.1 mm track, the finished track width ranges from 0.08 and 0.12 mm.	
PTH annular ring	$\geq 0.20\text{mm}$	2-layer: 1 oz: Recommended 0.25 mm or above; absolute minimum 0.18 mm 2 oz: 0.254 mm or above Multilayer: 1 oz: Recommended 0.20 mm or above; absolute minimum 0.15 mm 2 oz: 0.254 mm or above	
NPTH pad annular ring	$\geq 0.45\text{mm}$	Recommended 0.45 mm or more. This is to allow a 0.2 mm ring of copper to be removed around the hole for the sealing film to attach. Pad sizes smaller than the recommended value can result in the annular ring being very thin or completely missing.	
BGA	0.2mm	① 0.2mm-0.25mm BGA pad diameter requires ENIG ② BGA pad to trace clearance $\geq 0.1\text{ mm}$ (min. 0.09 mm for multilayer boards) ③ Vias can be placed within BGA pads using filled and plated-over vias	

Trace coils	0.15/0.15mm	Minimum trace width/clearance: 0.15/0.15mm, when traces are covered by solder mask (1oz). Minimum trace width/clearance: 0.25/0.25mm, when traces are NOT covered by solder mask (1oz). ENIG only (high risk of short circuit with HASL)
Hatched grid width and spacing	0.25 mm	
Same-net track spacing	0.25mm	
Inner layer via hole to copper clearance	0.2mm	
Inner layer PTH pad hole to copper clearance	0.3mm	
Pad to track clearance	0.1mm	Min. 0.1 mm (stay well above if possible). Min. 0.09 mm locally for BGA pads
SMD pad to pad clearance (different nets)	0.15mm	More details of SMD pad spacing: SMD Components Minimum Spacing Minimum SMD pad: 0.25mm × 0.25mm
Via hole to Track	0.2mm	
PTH to Track	0.28mm	0.35mm is recommended, minimum 0.28mm
NPTH to Track	0.2mm	

Soldermask

Features	Capability	Description	Patterns
Soldermask Expansion	1:1	LDI equipment upgraded in June 2025. Pad size:soldermask opening can be 1:1 (Previous production file will be followed in Reorder). Keep at least 0.09 mm clearance between soldermask openings and neighboring traces.	
Soldermask bridge	0.10mm	1oz: Min. pad spacing: 0.10 mm (green, red, yellow, blue, purple) Min. pad spacing: 0.13 mm (black, white) 2oz: Min. pad spacing: 0.20 mm (any color)	
Plugged vias	Filled with soldermask	Vias are filled with soldermask for an opaque finish. Click for detailed explanation ① Filled vias must not have soldermask openings on either side ② Filled vias should have ≥ 0.35 mm clearance from other soldermask openings (e.g. pads) ③ Filled vias must be no wider than 0.5 mm diameter	
JLCPCB Via-in-Pad Process	Epoxy Filled & Capped Copper paste Filled&Capped	Vias are filled with epoxy resin or copper paste and then plated over to achieve an opaque and smooth finish. Click for detailed explanation ① Vias are filled and plated over. Choose copper paste filling for applications requiring high thermal conductivity. ② This process is the default for 6-layer and above multilayer boards. ③ Compatible with via diameters from 0.15 to 0.55 mm.	

Solder mask dielectric constant	3.8	
Solder mask ink thickness	$\geq 10\mu\text{m}$	

Legend

Features	Capability	Description	Patterns
Minimum Line Width	$\geq 0.15\text{mm}$	Characters width less than 0.15mm will be unidentifiable.	
Minimum text height	40 mil (1.0mm)	Characters height less than 40 mil(1.0mm) will be unidentifiable.	
Character width to height ratio	1:6	The preferred ratio of width to height is 1:6.	
Hollow-carved Character width to height ratio	1:6	The preferred ratio of width to height is 1:6	
Pad To Silkscreen	0.15mm	The Minimum Distance Between Pad and Silkscreen is 0.15mm.	

Outline

Features	Capability	Description	Patterns
Routed	0.2mm	① Copper clearance from routed board edges: $\geq 0.2\text{ mm}$ ② Copper clearance from routed slots: $\geq 0.2\text{ mm}$ ③ Dimension tolerance for routed board edges: $\pm 0.2\text{ mm}$ (regular precision); $\pm 0.1\text{ mm}$ (high precision) ④ Minimum dimension 50*50mm for high precision, and at least 3 tooling holes with minimum 1.5mm diameter on different corners. ⑤ Minimum slot width for aluminum/copper core PCB: 1.6mm.	
V-Cut	0.4mm	① Copper clearance from V-cut board edges: $\geq 0.4\text{ mm}$ ② Dimension tolerance for V-cut board edges: $\pm 0.4\text{ mm}$. PCB thickness $\geq 0.6\text{ mm}$ ③ Zero panel board spacing by default. Alternatively, V-cut along one direction with no spacing and route along the other direction with 1.6 or 2 mm board spacing. ④ Min. panel dimensions: 70 × 70 mm; max. panel dimensions: 475 × 475 mm ⑤ V-cut groove angle: 25° ⑥ Minimum spacing between two v-cuts: 2mm(3mm is recommended)	
Mouse bites Panel	0.2mm	① Copper clearance from non-mouse-bite board edges: $\geq 0.2\text{ mm}$ ② Dimension tolerance for non-mouse-bite board edges: $\pm 0.2\text{ mm}$ (regular precision); $\pm 0.1\text{ mm}$ (high precision) ③ Panel board spacing: 1.6 or 2 mm ④ Serrated edges will remain after depanelization ⑤ Minimum tooling edge width: 3 mm. For SMT assembly at JLCPCB, use 5 mm tooling edges, 2 mm tooling holes, and 1 mm fiducials centered at 3.85 mm from the panel edges. ⑥ Recommended diameter of mouse bite is 0.5mm-0.8mm; Recommended distance between the two mouse-bites is 0.2-0.3mm. The minimum width of breakaway tab is 4mm. For breakaway with mouse-bites, the minimum width is 5mm.	
Panelization with space	2mm	The spacing between boards should be $\geq 2\text{ mm}$, as narrow spacing results in difficulties for routing and V-cut.	
Panel of Circular PCBs	$\geq 20\text{mm} \times 20\text{mm}$	The single round board size should be $\geq 20\text{mm} \times 20\text{mm}$ when choose panel by JLCPCB. Panelize with stamp holes and add tooling strips on four board edges	

FAQs about Rigid PCB Capabilities

What is the maximum number of layers supported by JLCPCB for standard production?

JLCPCB supports up to 32 layers in standard production. Layers from 1 to 6 offer the optimal balance between cost, lead time, and reliability. For designs exceeding 10 layers, we recommend submitting the stack-up diagram for engineering review prior to ordering to ensure manufacturability and controlled impedance performance.

Does JLCPCB provide impedance control, and what tolerance can be achieved?

Yes, JLCPCB offers impedance control for multi-layer boards. The standard tolerance is $\pm 10\%$. Tighter tolerance of $\pm 5\%$ is available upon special request, which requires controlled dielectric materials, precise stack-up design, and may involve additional verification.

What are the key differences in manufacturing capabilities between standard and advanced options?

Standard options include mechanical drilling, regular trace/space, and common surface finishes with faster lead times. Advanced options such as blind/buried vias, HDI (laser vias), heavy copper (up to 3oz), and tight impedance control provide higher design density and performance but typically require DFM review and may increase both cost and production time.

Which factors most significantly affect PCB board thickness and how can it be controlled?

Board thickness is primarily determined by the number of layers, copper weight, core and prepreg thickness, and surface finish. JLCPCB offers standard thicknesses from 0.4 mm to 3.2 mm. We recommend specifying the desired finished thickness in the order notes or stack-up drawing so our engineers can optimize the build-up accordingly.

How does panelization affect pricing and manufacturing quality?

Proper panelization significantly reduces unit cost by maximizing material utilization and improving production efficiency. JLCPCB supports V-Scoring and Stamp Hole (Mouse Bite) panelization. We recommend maintaining a minimum 2 mm spacing between boards and 5–8 mm handling rails to ensure mechanical stability during assembly and shipping.

Rigid PCBs to Your Specs

Upload your rigid board Gerber files now to experience industry-leading production speed and certified quality control.

Online Chat >

Chat with our live agent for fast reply.

Mon-Fri: 24 hours, Sat: 9am-6pm, GMT+8

Email Us >

Contact us at support@jlcpcb.com

Typically reply within hours.

Help Center >

Get instant answers.

24/7 Available.

Products

FR-4 PCBs

Flexible PCBs

Metal Core PCBs

High-Frequency PCBs

PCBA Service

PCB Layout

SMT Stencil

Flex Heater

Support

Help Center

Contact Us

Community

Shipping & Delivery

Payment Methods

How to Order

How to Track

After-Sales Service

Blog

Company

About Us

Quality Assurance

How We Work

Certifications

Security

Environment

JLCAPI

News

Cooperation

Electronics

EasyEDA

JLCPCB

JLCSFM

LCSC

OSHWLAB

Mechanical

JLC3DP

JLCCNC

JLCMC

CONNECT WITH US

© 2026 JLCPCB.COM All Rights Reserved. Privacy Policy Terms & Conditions Cookies Policy

- [Home](#)
- [Cart](#)
- [Files](#)
- [Message](#)
- [Me](#)